

Project Initialization and Planning Phase

Date	15 July 2026
Team ID	XXXXX
Project Name	FloodGuard - Flood Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This proposal outlines an ML-powered flood prediction system that converts rainfall, temperature, humidity and cloud-cover data into an instant, probability-scored flood risk assessment, delivered through a clean, responsive Flask web application.

Project Overview

Field	Details
Objective	Build and deploy an ML system that predicts flood likelihood from weather data, with a usable web interface for non-technical users
Scope	Covers the full pipeline: data preprocessing, model training/comparison, Flask backend, responsive frontend, database logging, and cloud deployment

Problem Statement

Field	Details
Description	Raw weather bulletins do not translate into a clear flood/no-flood signal, leaving residents and officers to interpret risk manually and often too late
Impact	Delayed or unclear warnings increase risk to life and property; a data-driven early-warning tool improves preparedness and response time

Proposed Solution

Field	Details
Approach	Train and compare multiple classifiers (Decision Tree, Random Forest, KNN, XGBoost) on historical weather/flood data; automatically select the best-performing model by F1-score and serve it through a Flask API
Key Features	- Real-time flood probability prediction via a web form - Automatic model comparison and selection during training - Prediction history log backed by a relational database - Responsive, modern UI with dynamic (AJAX) result rendering

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware - Computing	CPU specification for training/serving	Standard cloud CPU instance (Render free-tier web service)
Hardware - Memory	RAM specification	512 MB (Render free-tier instance)
Software - Framework	Web framework	Flask 3.1
Software - Libraries	ML & data libraries	scikit-learn, XGBoost, pandas, NumPy, joblib, matplotlib, seaborn

Software - Dev Environment	IDE / editor	VS Code / PyCharm
Data	Source, size, format	flood_dataset.xlsx, 115 records, 10 weather features + target